# TPG4190 Seismic data acquisition and processing
# Lecture 9: Processing

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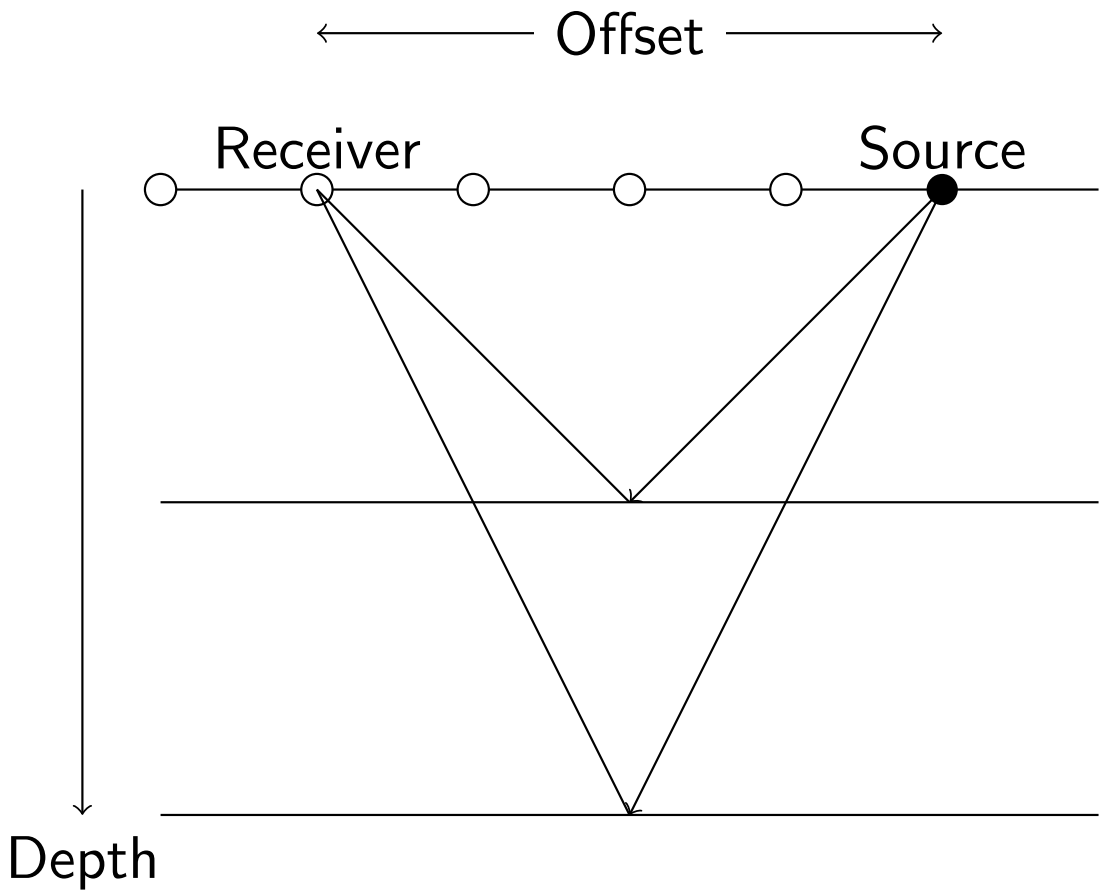
# Overview

- Filters
- Basic processing sequence
- Basic+ processing sequence
- Basic++ processing sequence

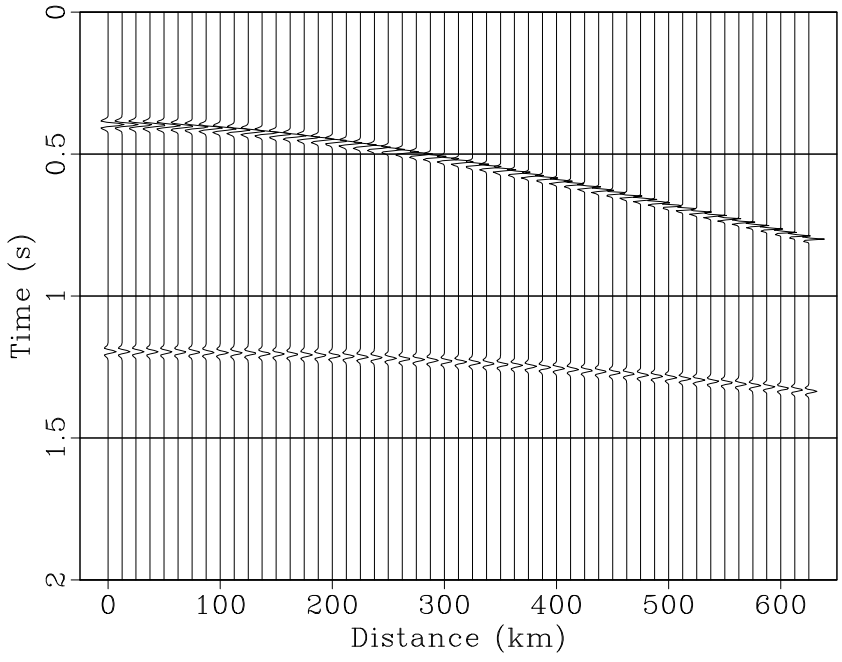
# Basic processing sequence

1. Input data
2. Velocity analysis
3. NMO + Stack
4. Output result

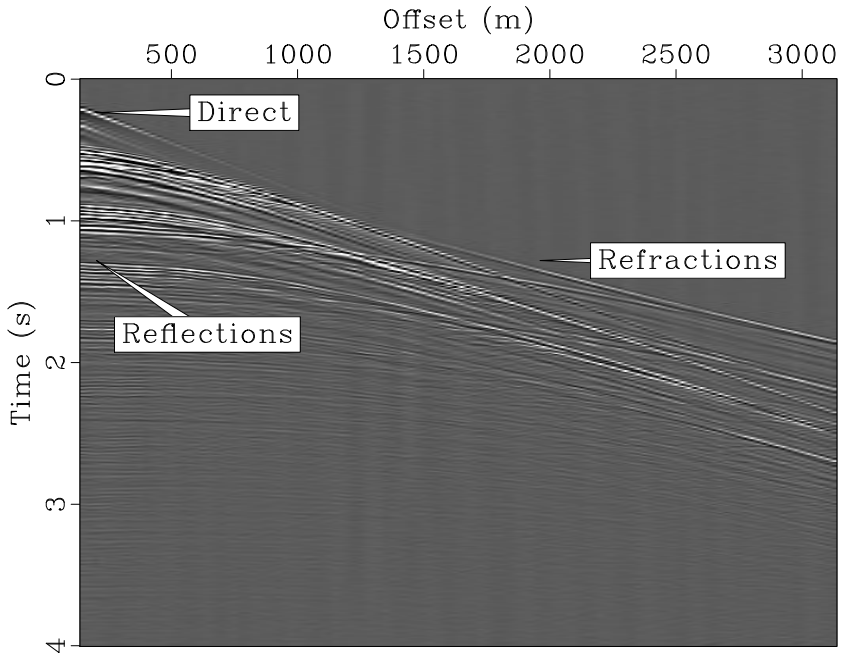
# Seismic marine data acquisition



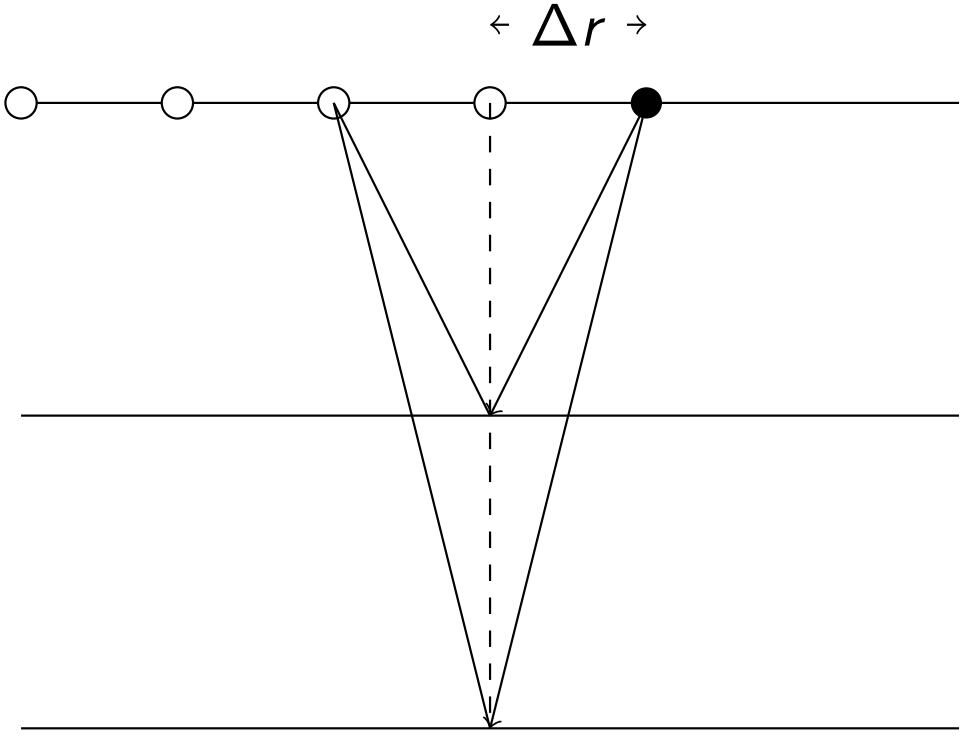
# Schematic shot record



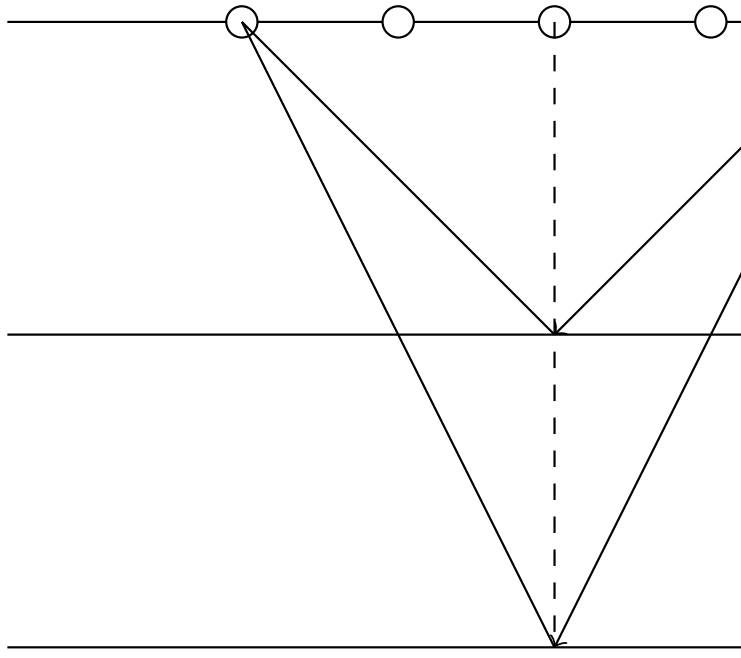
# Real shot record



# The cmp method

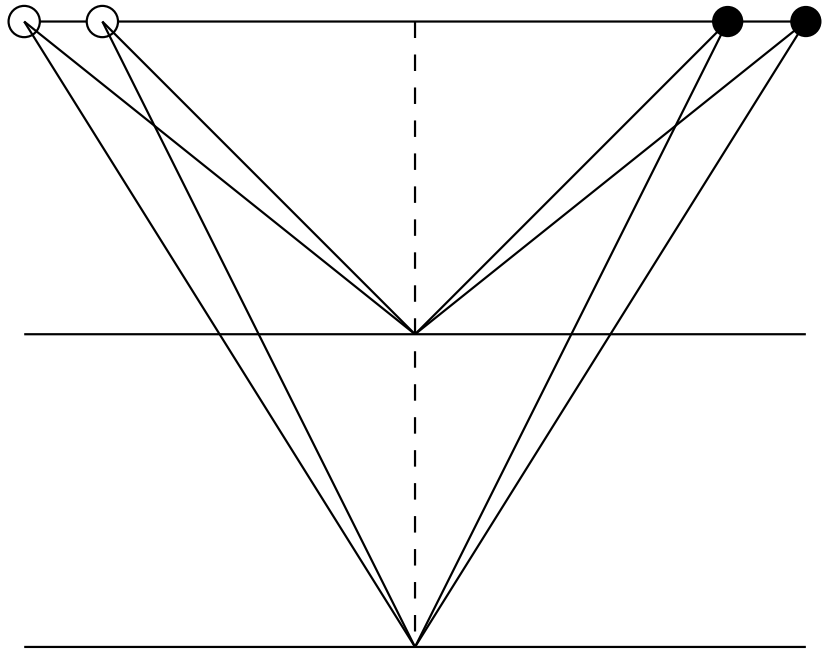



Common Midpoint(Cmp)



Common Midpoint(

# Common midpoint (cmp) gather



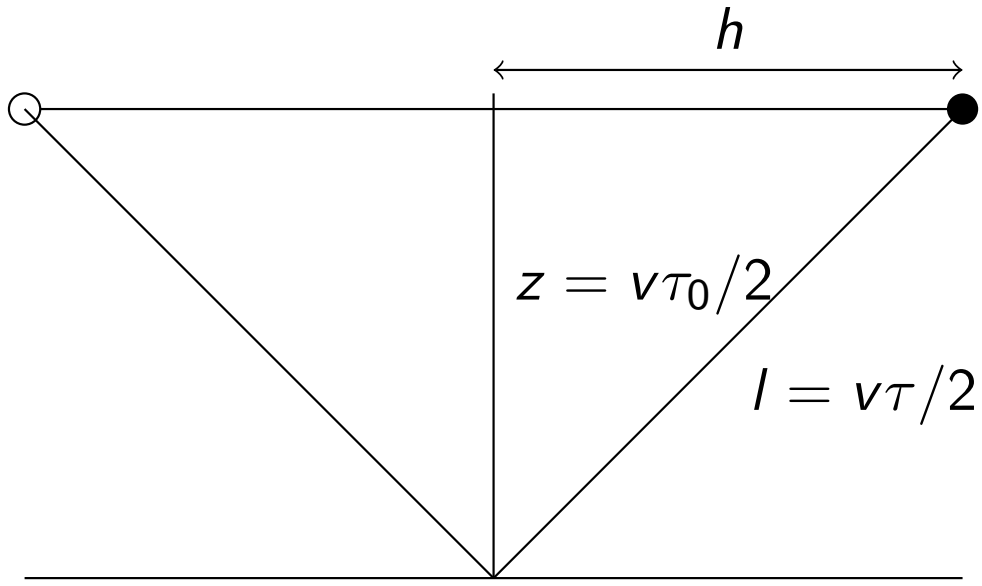
# Midpoint and Offset Coordinates

$$x_m = \frac{s+r}{2}, \tag{1}$$

$$h = \frac{s-r}{2}, \tag{2}$$

- $x_m$: Midpoint coordinate
- $h$: Offset
- $s$: Source coordinate
- $r$: Receiver coordinate

# NMO and Stack




The traveltime $\tau(h)$ is:

$$l^2 = z^2 + h^2 \tag{3}$$

which gives by inserting $v\tau/2$ for $l$ and $v\tau_0/2$ for $z$

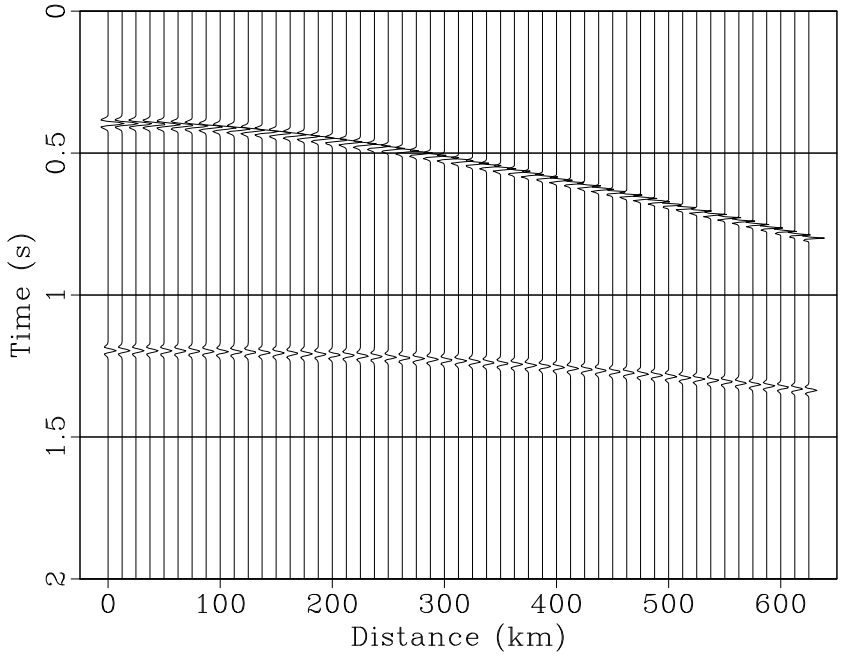
$$\tau(h) = \sqrt{t_0^2 + 4h^2/v^2}. \tag{4}$$

# NMO and Stack

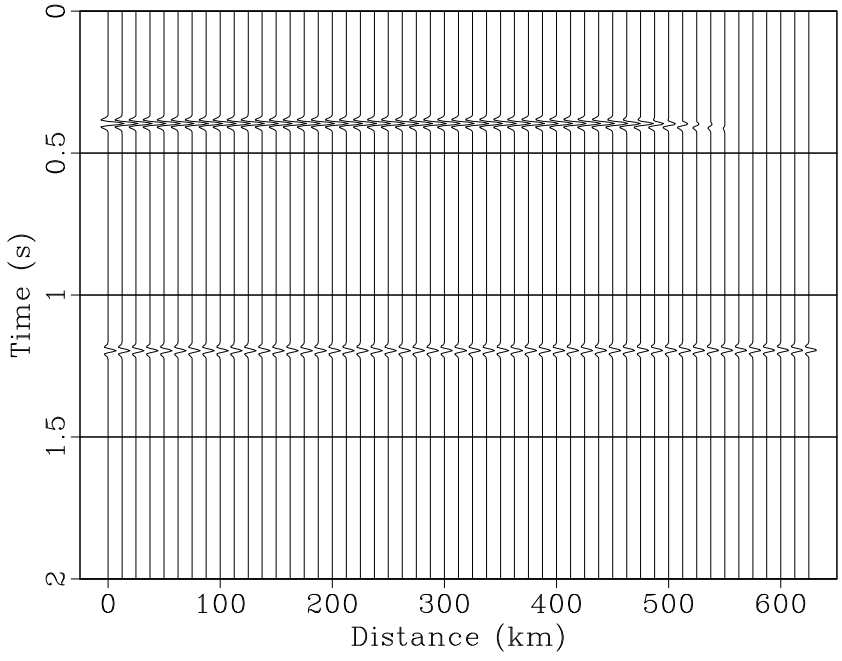
Nmo-correction:

$$\Delta\tau = \tau(h) - \tau_0, \tag{5}$$

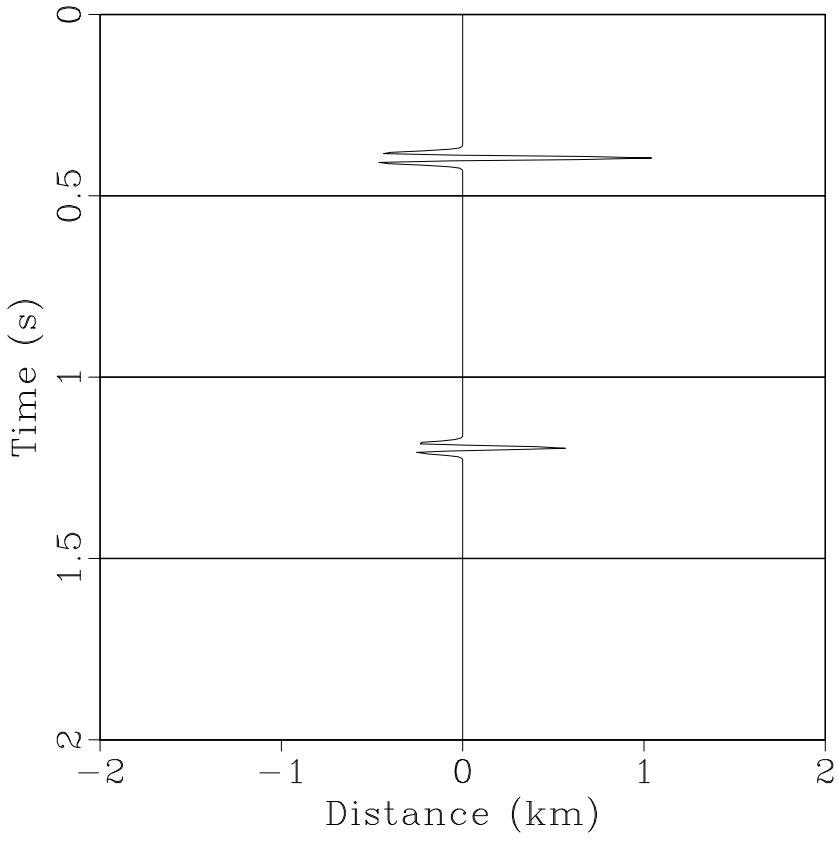
# Cmp



# Nmo



# Stack



# NMO and Stack

Average velocity $v_{rms}$ defined by

$$v_{rms}^2(t_0) = \frac{1}{t_0} \int_0^{t_0} v^2(t) dt \tag{6}$$

$v(t)$ : Interval velocity. The traveltime equation (4) then becomes

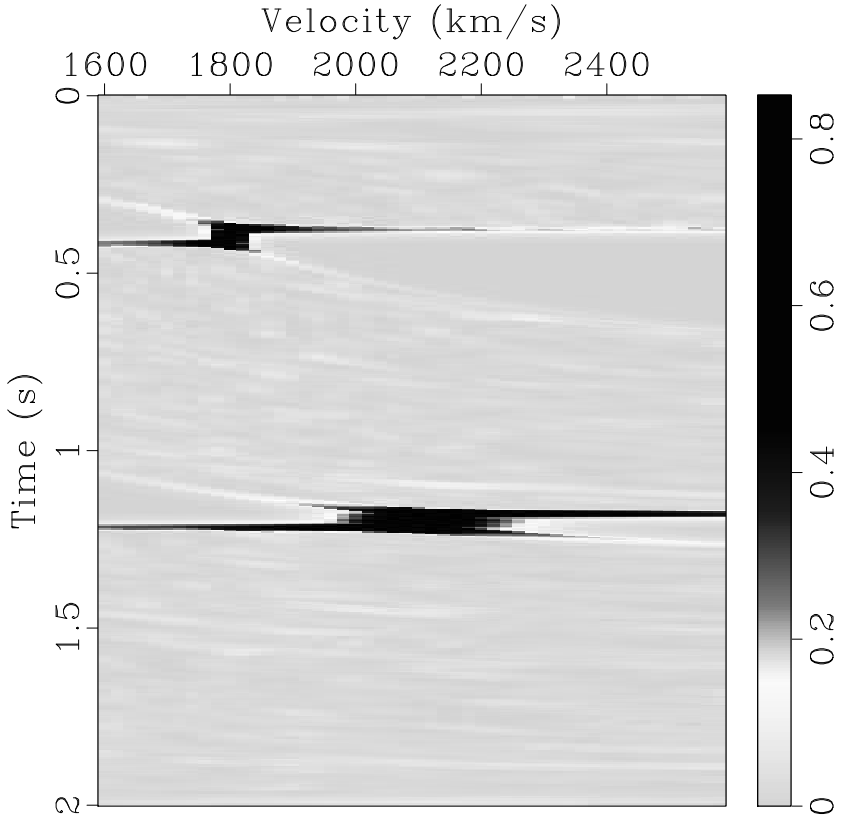
$$\tau(h) = \sqrt{t_0^2 + 4h^2 / v_{rms}^2(t_0)}. \tag{7}$$

# Velocity analysis

Estimate $v_{rms}$ from equation (4)

1. Make a guess of $v$
2. Perform Nmo correction for all $t_0$ with using guess of $v$
3. Make a stack trace
4. Perform above steps for a range of guesses of $v$
5. Plot all stack traces in a velocity spectrum

# Velocity spectrum



# Semblance

Stack is usually replaced with semblance to get better velocity spectra

$$S(t) = \frac{\int_0^{h_{max}} p^2(t,h)}{\int dt_0^T \int_0^{h_{max}} p^2(t,h)} \tag{8}$$

- $p(t,h)$ : data
- $h_{max}$: maximum offset.

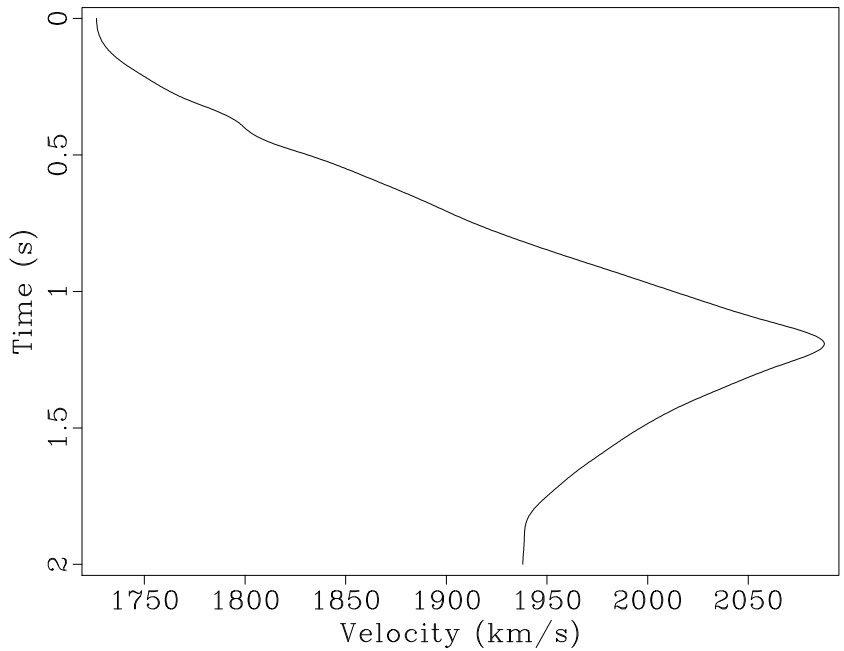
This equation can not be used directly. Why?

# Semblance

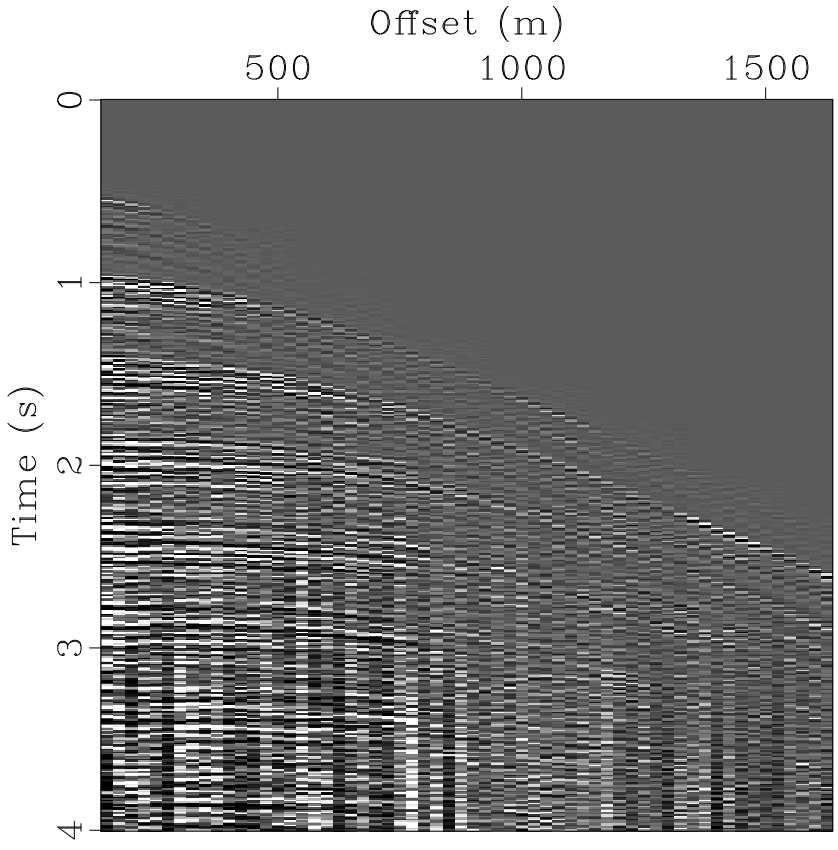
$$S_k = \frac{\sum_{i=0}^{N_h} p_{k,i}^2}{\sum_{k=0}^{N_t} \sum_{i=0}^{N_h} p_{k,i}^2} \tag{9}$$

- $S_k = S(t = k\Delta t), k = 0, \ldots, N_t$
- $p_{k,i} = p(t = k\Delta t, h = i\Delta h), i = 0, \ldots, N_h$.
- $\Delta t$ time sampling interval,
- $\Delta h$ distance between offsets
- $N_t$,$N_h$ : No of time samples and No of offsets

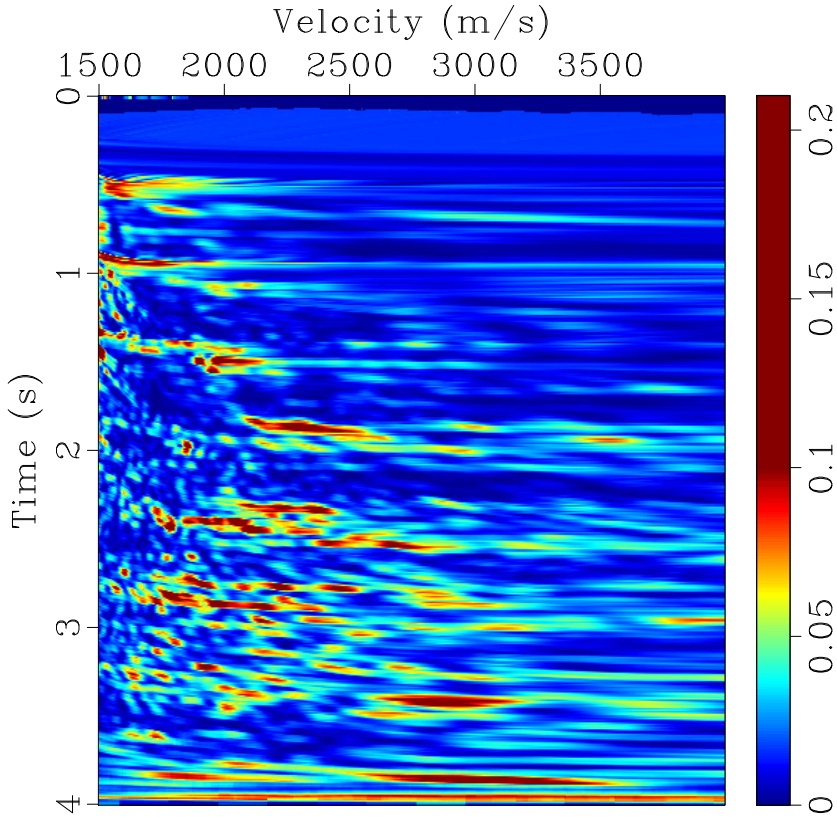
# Velocity analysis



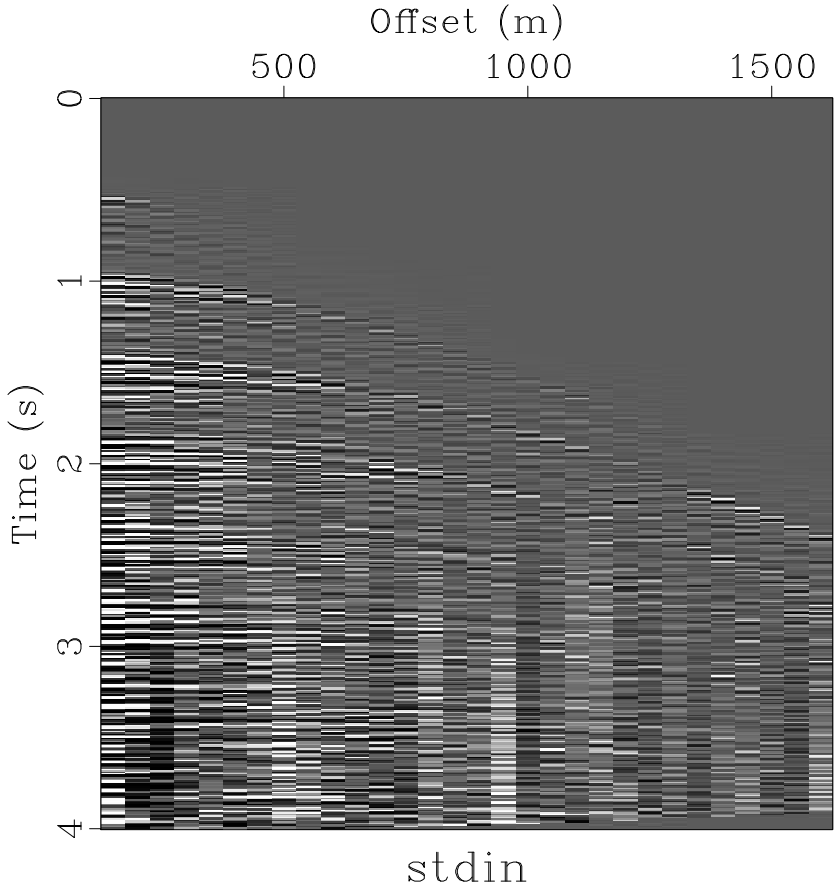
# Velocity analysis



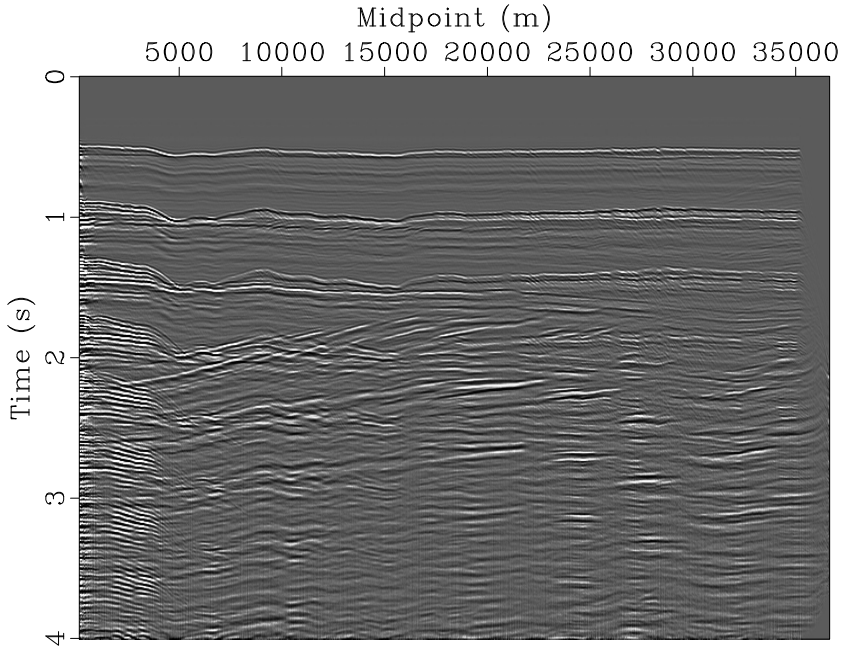
# Velocity analysis



# Velocity analysis



# Velocity analysis



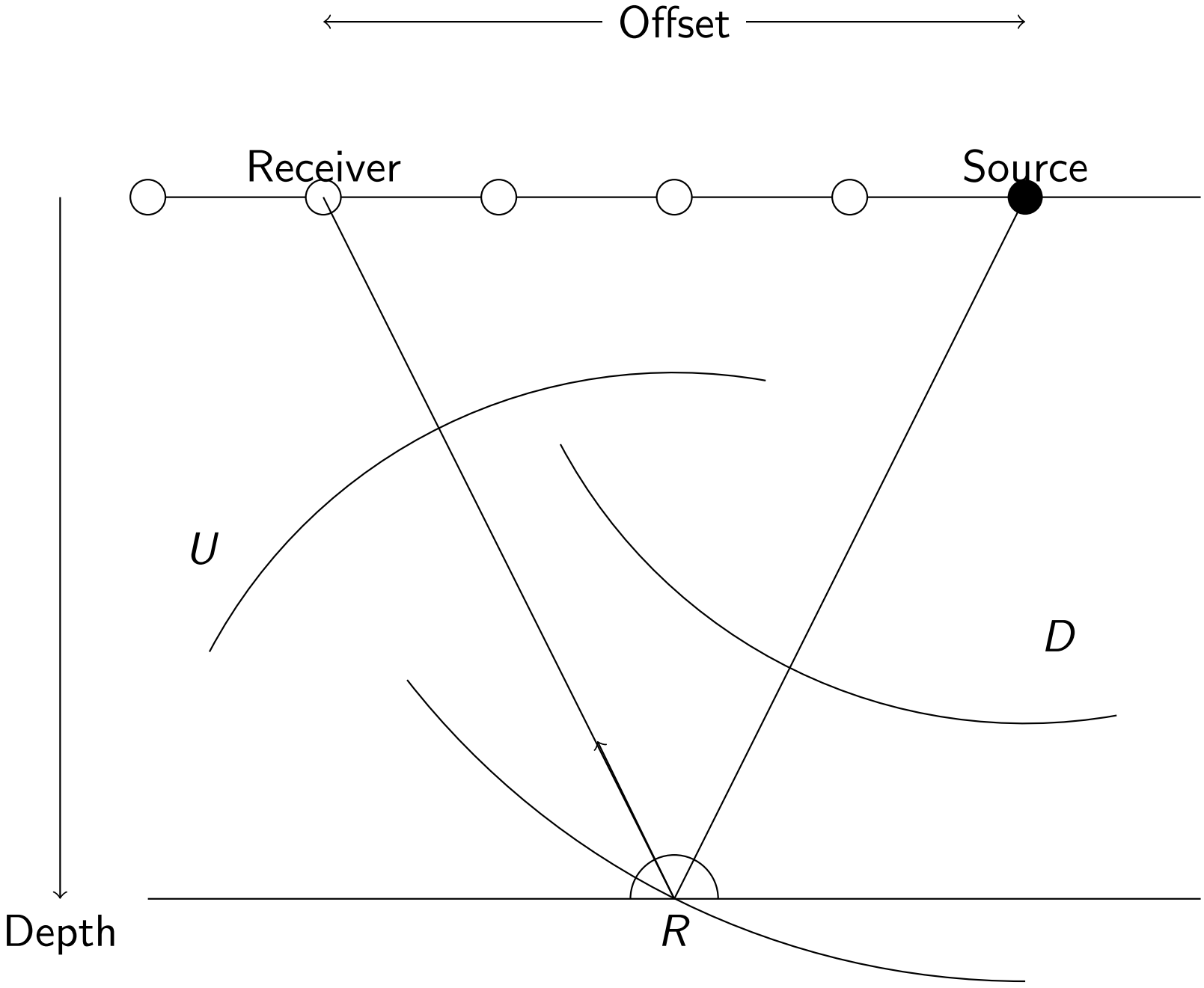
# Summary

- Seismic data acquisition
- CMP method
- NMO-correction
- Velocity analysis

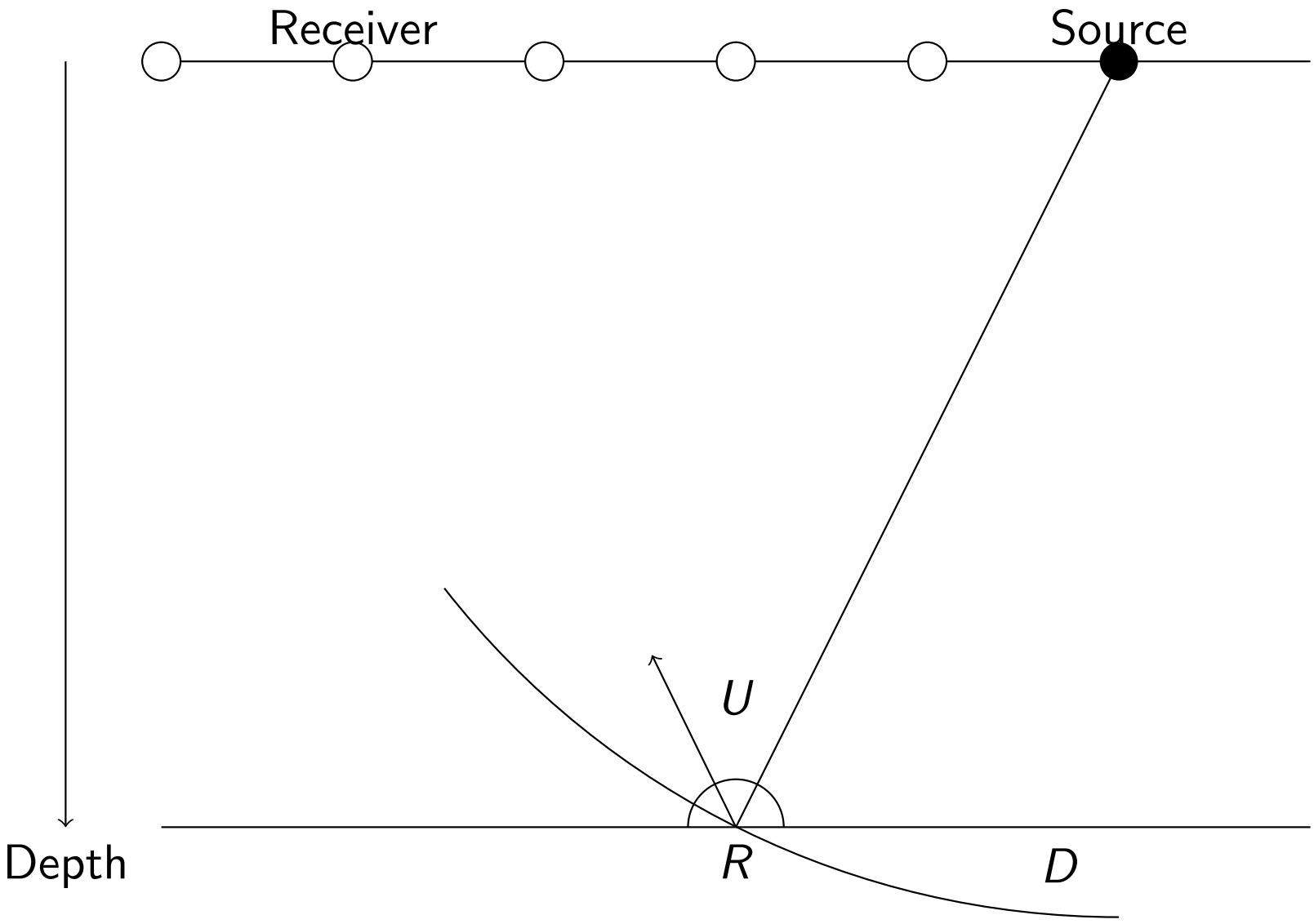
# Overview

- Migration principles
- Kirchhoff Time migration

# Migration principles



# Migration principles



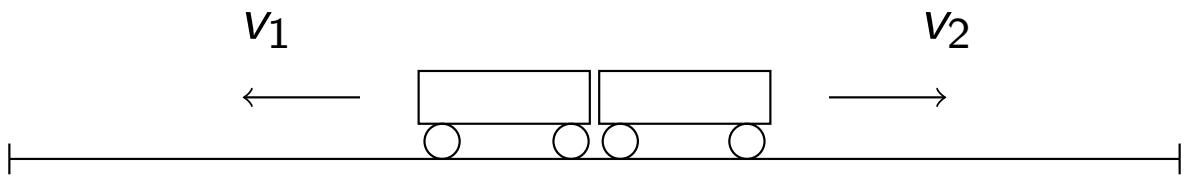
# Migration principles

Make an image by cross-correlating the downgoing and upgoing waves

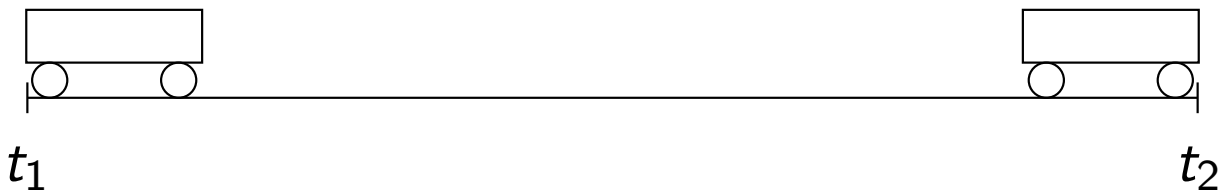
$$R(\mathbf{x}) = \int dt U(\mathbf{x}, t) D(\mathbf{x}, t), \tag{10}$$

where $R$ is the reflectivity and $\mathbf{x} = (x, y, z)$ denotes a position in space, while $t$ is the time.

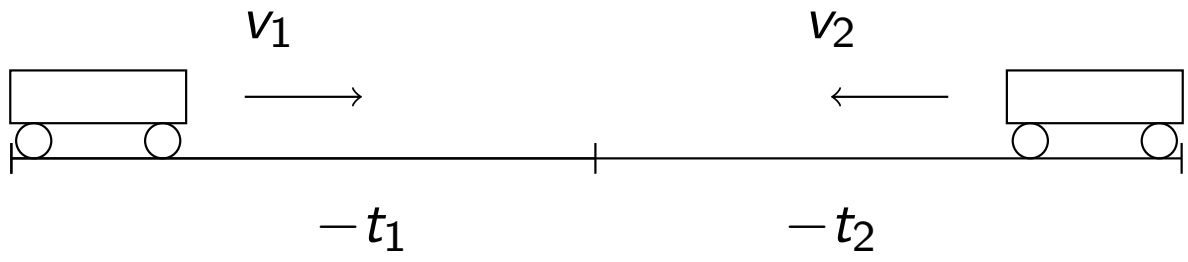
# Time reversal 1



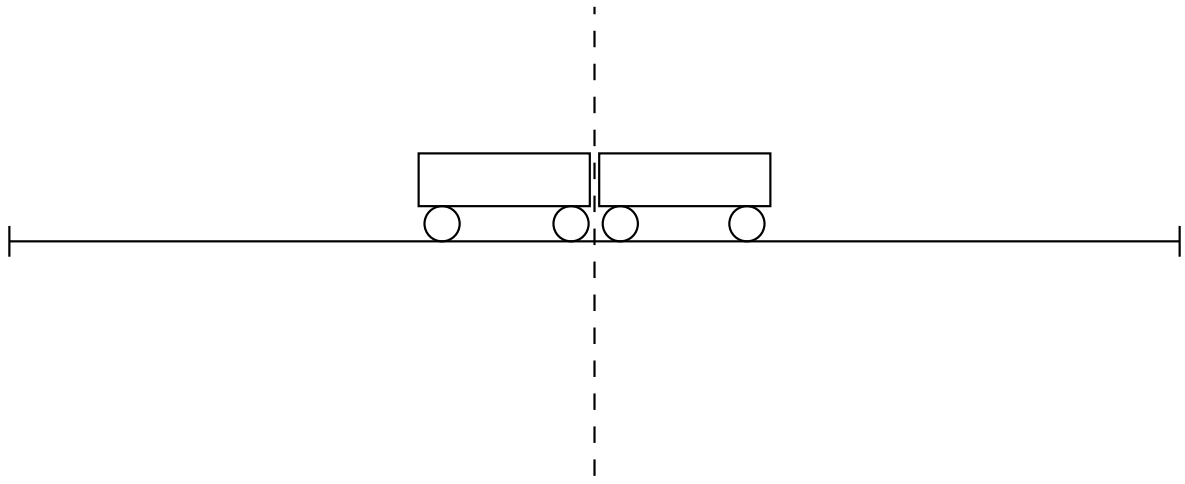
# Time reversal 2



# Time reversal 3



# Time reversal



# Seismic imaging

Forward modeling

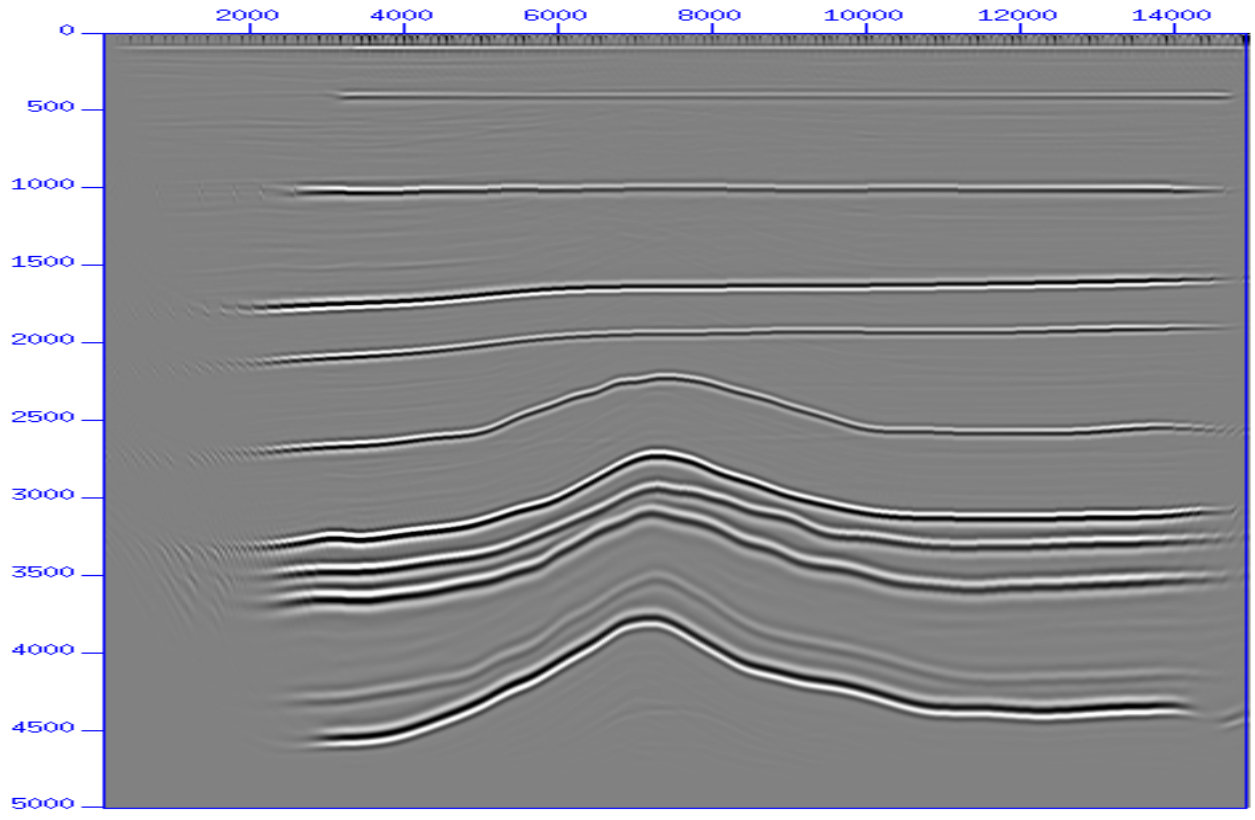
# Time reverse modeling

Backward modeling

# Cross correlation

Imaging

# Complete image



# Kirchhoff migration

Simplest approach for up- and downgoing waves

$$D(\mathbf{x}, t) = A\delta(t - \tau_s),$$
$$U(\mathbf{x}, t) = BP(t + \tau_r), \tag{11}$$

where

- $\mathbf{x}$: reflection point,
- $A$ and $B$: amplitude factors
- $P$: Data at the surface
- $\tau_s$ traveltime from the source to the reflection point
- $\tau_s$ traveltime from the receiver to the reflection point

# Kirchhoff migration

Using equation (11) and equation (10), I get

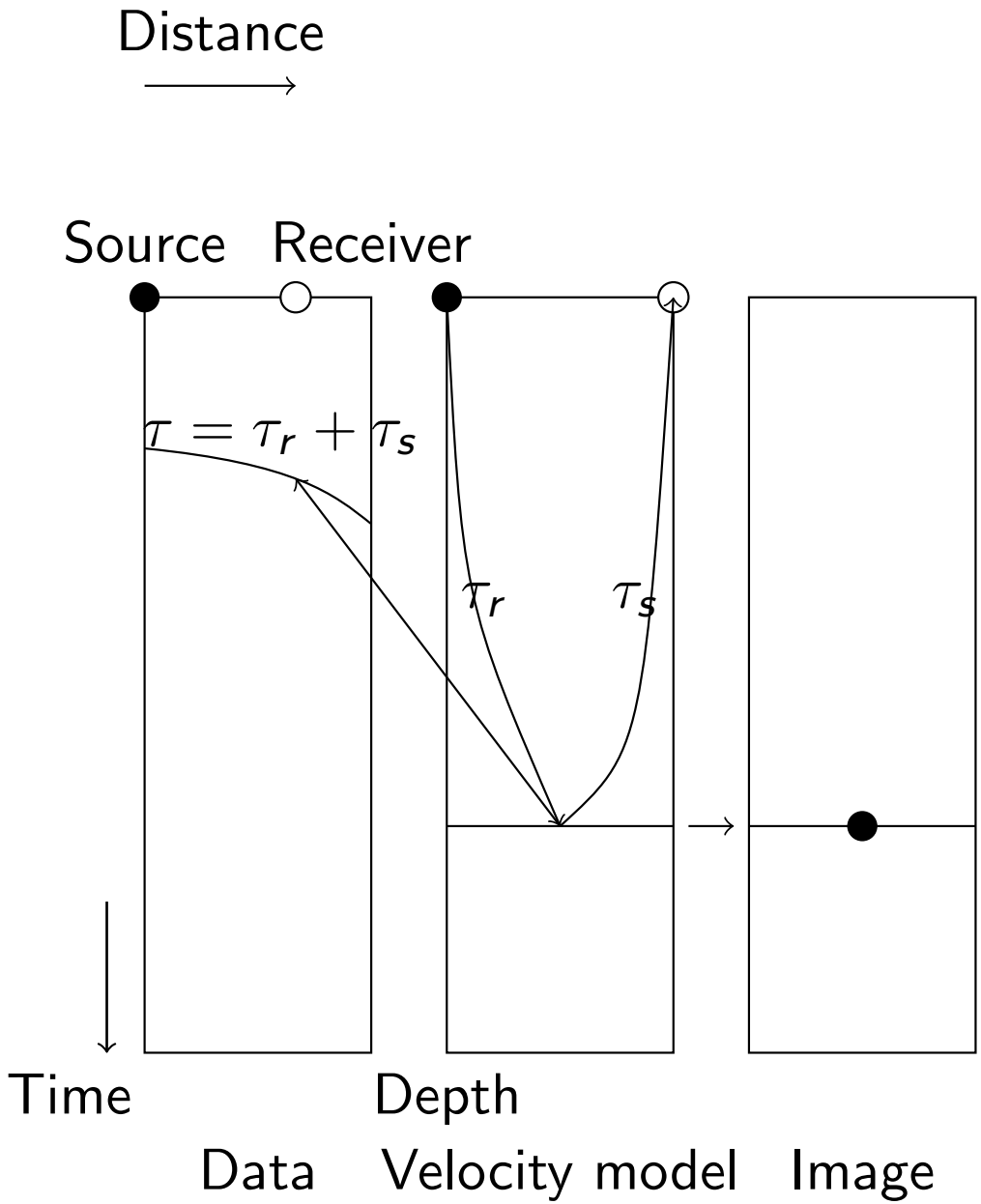
$$R(\mathbf{x}) = \int dt A\delta(t - \tau_s)BP(t + \tau_r), \tag{12}$$

which after integration gives

$$R(\mathbf{x}) = ABP(\tau_s + \tau_r). \tag{13}$$

If we disregard the amplitude factor $AB$ the image is simply equal to the amplitude of the recorded data at a time equal to $\tau = \tau_r + \tau_s$.

# Kirchhoff migration



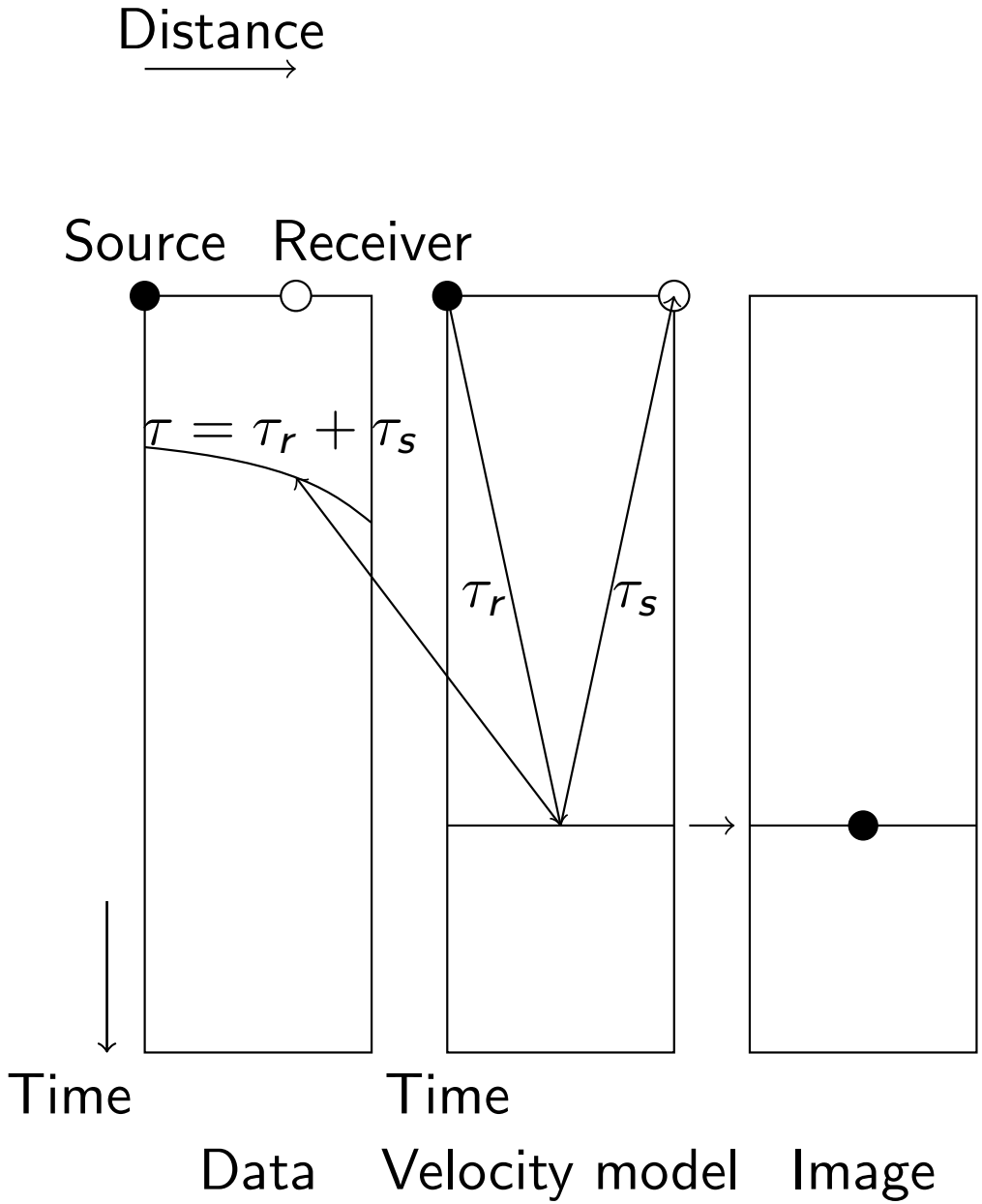
# Kirchhoff migration

Many source-receiver pairs will contribute to the same imaging point, so that equation (13) becomes

$$R(\mathbf{x}) = \sum_{r,s} P(\tau_s + \tau_s), \tag{14}$$

where we have neglected the amplitude factors and the summation is over all source-receiver pairs contributing to the image point.

# Kirchhoff migration



# Kirchhoff migration

Assume velocity $c(\mathbf{x})$ is depth dependent only, $c = c(z)$, then

$$\tau_s = \sqrt{\frac{(x - x_s)^2 + (y - y_s)^2}{c^2} + \tau_0^2},$$
$$\tau_r = \sqrt{\frac{(x - x_r)^2 + (y - y_r)^2}{c^2} + \tau_0^2}, \tag{15}$$

- $(x_r, y_r)$, $(x_s, ys)$: Source, receiver positions
- $\tau_0 = z/c$ :vertical traveltime and
- $c$ : velocity.

# Kirchhoff migration

We then get for the total traveltime $\tau$

$$\tau = \sqrt{\frac{(x - x_s)^2 + (y - y_s)^2}{c^2} + \tau_0^2} + \sqrt{\frac{(x - x_r)^2 + (y - y_r)^2}{c^2} + \tau_0^2}. \quad (16)$$

The image $R$ is then

$$R(x, y, \tau_0) = \sum_{r,s} P(\tau_s + \tau_s), \quad (17)$$

# Zero-offset simple migration

We get for zero-offset (stack) data (2D)

$$\tau = 2\sqrt{\frac{(x - x_s)^2}{c^2} + \tau_0^2}. \tag{18}$$

# Zero-offset simple migration

Migration of zero-offset section can then be done as:

- Chose a point $x_s, \tau_0$ on the section.
- Loop over all $x$ values, compute $\tau$ and sum all $P(\tau)$ (Summing over an hyperbola with center at $x_s, \tau_0$.)
- Put the sum at location $x_s, \tau_0$ in the output.
- Repeat for all possible points $x_s, \tau_0$.

# Basic+ processing sequence

1. Input data
2. Velocity analysis
3. NMO + Stack
4. Zero-offset (poststack) migration,
5. Output result

Nobody with full possession of their faculties uses a processing sequence like this today, except for initial QC.

# Basic++ processing sequence (time)

1. Input data
2. preprocessing (designature, debubble, etc..)
3. Multiple removal
4. Initial prestack migration
5. Velocity analysis on demigrated data.
6. Final prestack migration
7. Multiple removal
8. Stack
9. Output result